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Redefining Reading: How Digital Tools are Changing Functional Literacy

This poster examines how digital technologies, including text-to-speech, voice assistants, and generative AI, are fundamentally reshaping traditional definitions of functional literacy in the digital age. The core objective is to determine how youth and teen readers, particularly those in under-resourced communities, utilize these tools to access information and engage with text, thereby challenging institutional definitions of reading. The study employed a mixed-methods approach, combining a theoretical historical review of scholarly literature with a local empirical investigation. The empirical component included literacy data analysis and a survey of 55 students in grades 6-12 exploring their digital tool usage and motivations for using those tools for reading and writing. Analysis of assessment data revealed significant literacy deficits among the target student population. Scholarly research confirms that text-to-speech (TTS) technology significantly enhances reading comprehension for struggling readers by reducing cognitive load. However, survey results indicate a "digital disconnect," with youth reporting more frequent use of generative AI and voice assistants than of dedicated TTS applications. Students' primary motivations for using these tools for reading focus on functional performance over explicit comprehension. This research demonstrates a necessary institutional shift: the definition of functional literacy must be expanded to validate multimodal digital supports as legitimate, equitable pathways to knowledge and participation. This validation would empower struggling, under-resourced youth by recognizing compensatory strategies like TTS as essential tools for achieving inclusion and academic success.